



Dual response in conjoint analysis

Cheng-Yu Hung¹ · YiChun Miriam Liu² · Jeff D. Brazell³ · Greg M. Allenby¹

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Abstract

Conjoint analysis often incorporates a no-choice option to allow respondents to opt out of a choice task. This paper examines the benefits of a dual-response format in which respondents first select their preferred alternative from among a set of available alternatives, excluding the no-choice option, and then decide whether they would actually buy it. We show that respondents employ a budget constraint when responding to the second response task but not to the first, indicating that respondents assess affordability separately from preference. We also show that our model with the dual-response format improves parameter estimation efficiency by 30% as measured by the Fisher information matrix relative to a standard model using a single-response format. We use our proposed budget constraint model to examine optimal pricing in a competitive market, where the standard model is shown to overestimate equilibrium prices.

Keywords Economic choice · Budget constraints · Competitive pricing · Nash price equilibrium

1 Introduction

Conjoint choice experiments, in which participants select their preferred option from one or more sets of alternatives, are widely used in marketing to measure consumer preferences for products and services. In conjoint choice analysis, including a constant

✉ Cheng-Yu Hung
hung.253@osu.edu

YiChun Miriam Liu
ycliu@iastate.edu

Jeff D. Brazell
jeff.brazell@eccles.utah.edu

Greg M. Allenby
allenby.1@osu.edu

¹ Ohio State University, Columbus, USA

² Iowa State University, Ames, USA

³ University of Utah, Salt Lake City, USA

or no-choice option—such as “none of these”—allows respondents to indicate that they do not prefer any of the presented options and would rather opt out of the choice task to save money. Adding a no-choice option to the set of choices is shown to better reflect real-life purchasing scenarios (Brazell et al., 2006), improve market share and volume predictions from analysis (Carson et al., 1994), eliminate statistical biases (Haaijer et al., 2001), and increase design efficiency (Anderson and Wiley, 1992).

We examine the benefit of using a dual-response choice format for the no-choice option, in which respondents first indicate their preferred choice alternative and then state whether they would actually purchase the chosen alternative. The dual-response choice format helps respondents feel less pressure to say “no” in the second follow-up question, as they have already been helpful in the first question (Tourangeau et al., 2000). Asking respondents if they would actually purchase their preferred alternative as a follow-up question also slows their pace and encourages them to think carefully about whether it fits within their budget (Gorman et al., 1996; Blackorby and Russell, 1997; Deaton and Muellbauer, 1980). Importantly, while the dual-response choice format increases the frequency of selecting the no-choice option, the relative shares after accounting for this difference remain unaffected. Moreover, it does not significantly alter the estimated part-worths, ensuring that inferences about preferences remain consistent (Brazell et al., 2006; Diener et al., 2006).

The analysis of dual-response choice data remains unclear. Some researchers collapse the dual responses in each choice task into a single response by deleting the first response if the respondent selects the no-choice option in the second question. This collapsed approach diminishes the power of the dual-response format, as it fails to capture information about the differences among the alternatives when the no-choice option is chosen. Other researchers model the responses to the two questions independently, assuming that the respondents recall the utility of selecting alternatives without retaining any private information after the first question (Diener et al., 2006; SawtoothSoftware, 2016). In other words, the error term realizations for two questions are different. However, since the second response question is presented immediately after the first question, respondents are likely to recall their utility for the choice alternatives and decide whether to actually purchase their most preferred alternative using the same private information.

In this paper, we propose a model for conjoint tasks with dual responses in which a budget constraint is imposed on the second task but not the first. Our model assumes that respondents retain private information for choice alternatives across their dual responses so that error term realizations, not observed by the analyst, are the same in the two response tasks. Furthermore, we show that our proposed model fits the data better and leads to a 30% improvement in statistical information for parameter estimation relative to a model that collapses the two responses into a single response.

The remainder of this paper is structured as follows. First, we develop our proposed model for the dual-response format with a budget constraint in Sect. 2. We then describe the survey used to collect our data in Sect. 3. Estimates of parameters are reported in Sect. 4, followed by a discussion of efficiency comparisons of dual versus single-response formats, and optimal pricing in a competitive market scenario in Sect. 5.

2 Model development

The standard specification of a discrete choice model in conjoint analysis assumes J choice alternatives and uses a dummy-variable representation of product features, denoted by $x_{j1}, x_{j2}, \dots, x_{jL}$ for alternative j , where one of the attribute levels is selected as the null level of the attribute. We assume that the utility of selecting alternative j for the respondent is a linear combination of the part-worths plus an error term, such that:

$$U_j = \beta'x_j + \beta_p p_j + \varepsilon_j, \quad \varepsilon_j \sim EV(0, 1)$$

where $x_j = [x_{j1}, \dots, x_{jL}]'$ is the vector of attribute levels for alternative j , β represents the part-worths for the corresponding attribute levels, β_p denotes the marginal utility of money, and p_j is the price of alternative j . The error term is assumed to follow an extreme value distribution with a scale parameter of one. For statistical identification, we set the utility from selecting the no-choice option as:

$$U_{nc} = \varepsilon_{nc}, \quad \varepsilon_{nc} \sim EV(0, 1)$$

Let V_j be the deterministic part of the utility from selecting alternative j , where $V_j = \beta'x_j + \beta_p p_j$. Then, under this specification, the choice probability results in the following logit expression of choice:

$$\begin{aligned} Pr(j) &= \frac{\exp(V_j)}{1 + \sum_{k=1}^J \exp(V_k)} \\ Pr(nc) &= \frac{1}{1 + \sum_{k=1}^J \exp(V_k)} \end{aligned} \quad (1)$$

where “nc” indicates the no-choice alternative.

2.1 Dual responses

The dual-response format asks respondents to identify their most preferred alternative and then indicate if they would actually buy it. An approach to modeling dual-response data is to collapse the responses into a single response by ignoring the answers to the first response task if the no-choice option is selected. This results in the model specification in Eq. 1, and we refer to this model as the Standard Model of choice.

An alternative model is to assume the dual responses are associated with a fresh draw of the error term, which represents private information known to the respondent but not to the analyst. In this specification, the probability of selecting alternative “i” in the first response and the second response is:

$$Pr(i, \text{first response}) = \frac{\exp(V_i)}{\sum_{k=1}^J \exp(V_k)}$$

$$Pr(i, \text{second response}) = \frac{\exp(V_i)}{1 + \exp(V_i)} \quad (2)$$

In contrast, our proposed Unified Model assumes respondents use the same private information about the alternatives when making their selections in both the first and the second response tasks. Similar to the utility in the conjoint model, the utility of selecting alternative j for the respondent in both the first and the second response tasks is:

$$\begin{cases} U_j = V_j + \varepsilon_j, & \varepsilon_j \sim \text{EV}(0, 1) \\ U_{nc} = \varepsilon_{nc}, & \varepsilon_{nc} \sim \text{EV}(0, 1) \end{cases} \quad (3)$$

where we set the utility from selecting the no-choice option in the same way as in the Standard Model specification. The joint probability of selecting alternative j and selecting the purchase option in the second response task is then:

$$\begin{aligned} Pr(j, \text{purchase}) &= Pr(\{U_j > U_k \forall k \neq j\} \cap \{U_j > U_{nc}\}) \\ &= Pr(U_j > U_k \forall k \neq j, k \in \{1, \dots, J, nc\}) \\ &= \frac{\exp(V_j)}{1 + \sum_{k=1}^J \exp(V_k)} \end{aligned}$$

and the joint probability of selecting alternative j and selecting the no-choice option in the second response task is:

$$\begin{aligned} Pr(j, nc) &= Pr(\underbrace{\{U_j > U_k \forall k \neq j\}}_A \cap \underbrace{\{U_j < U_{nc}\}}_B) \\ &= Pr(\{U_j > U_k \forall k \neq j\}) - Pr(\{U_j > U_k \forall k \neq j\} \cap \{U_j > U_{nc}\}) \\ &= Pr(\{U_j > U_k \forall k \neq j\}) - Pr(j, \text{purchase}) \\ &= \frac{\exp(V_j)}{\sum_{k=1}^J \exp(V_k)} - \frac{\exp(V_j)}{1 + \sum_{k=1}^J \exp(V_k)} \\ &= \frac{\exp(V_j)}{\sum_{k=1}^J \exp(V_k)} \times \frac{1}{1 + \sum_{k=1}^J \exp(V_k)} \end{aligned}$$

where the derivation from the first line to the second line is illustrated in Fig. 1. The joint probability of the proposed Unified Model can be summarized as:

$$\begin{aligned} Pr(j, \text{purchase}) &= \frac{\exp(V_j)}{1 + \sum_{k=1}^J \exp(V_k)} \\ Pr(j, nc) &= \frac{\exp(V_j)}{\sum_{k=1}^J \exp(V_k)} \times \frac{1}{1 + \sum_{k=1}^J \exp(V_k)} \end{aligned} \quad (4)$$

Equation 4 is the same as that proposed by Diener et al. (2006) for their DR-AnyMax model using different assumptions. They distinguish two forms of dual response, one where the follow-up question is, “Would you actually buy the option you selected in

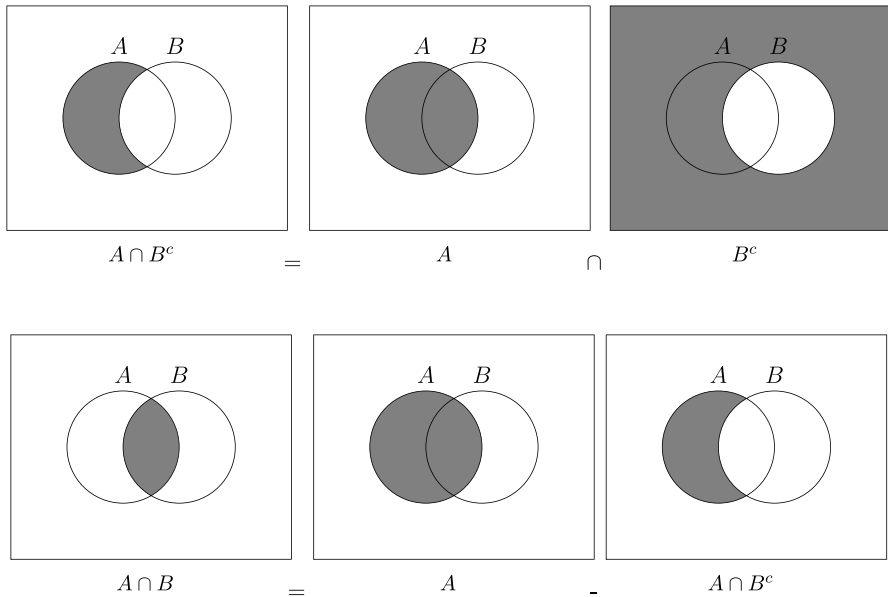


Fig. 1 Venn diagram illustration of choice probability

the previous question?”, which they refer to as DR-2Max and is equivalent to Eq. 2. Their other follow-up question for the DR-AnyMax model is, “Would you actually buy any of the options listed above?” They find that the DR-AnyMax model provides a better fit to dual-response data than the DR-2Max model.

2.2 Budget constraints

Budget constraints can be added to the model specification in the first response, second response, or both. We discuss the addition of the budget in the second question for illustration. Let C be the set of indices of alternative choices that satisfy the budget constraint, i.e., $C = \{j \mid p_j \leq M\}$, M is the monetary constraint, and the utility of selecting alternative j for the respondent in both the first and the second response tasks is the same as in Eq. 3. We assume that respondents disregard the budget constraint when making decisions in the forced choice task, but consider the budget constraint when deciding whether to actually purchase the selected alternative or not. The joint probability becomes:

$$Pr(j, \text{purchase}) = \begin{cases} \frac{\exp(V_j)}{1 + \sum_{k=1}^J \exp(V_k)}, & \text{if } p_j \leq M, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

$$Pr(j, \text{nc}) = \begin{cases} \frac{\exp(V_j)}{\sum_{k=1}^J \exp(V_k)} \times \frac{1}{1 + \sum_{k=1}^J \exp(V_k)}, & \text{if } p_j \leq M, \\ \frac{\exp(V_j)}{\sum_{k=1}^J \exp(V_k)}, & \text{otherwise.} \end{cases}$$

We restrict the price coefficient to be negative in our analysis using an algebraic transformation:

$$\beta_p = -\exp[\beta_p^*] \quad (6)$$

and estimate β_p^* unrestricted. The heterogeneity of the respondents is incorporated using a random-effects specification. For each respondent h , let β_h denote the part-worths for attribute levels, β_{ph}^* represent the transformed price coefficient, and M_h the budget constraint. Then, we have:

$$\phi_h = (\beta_h', \beta_{ph}^*, M_h)' \sim \text{Normal}(\bar{\phi}, \Sigma_\phi) \quad (7)$$

where $\bar{\phi}$ and Σ_ϕ are the mean and covariance matrix of the random-effects distribution.

3 Data

We examine the performance of the proposed model using data from respondents who participated in a conjoint survey for large-screen televisions (Liu et al., 2025). Participants were recruited from a United States panel (Amazon Mechanical Turk) for the study. To be included in the survey, respondents had to currently own a large-screen television, have participated in a recent large-screen television purchase, or intend to make a large-screen television purchase within the next six months. Qualified participants who successfully passed the screening questions were required to watch an instructional video explaining the choice task and defining the product attributes and their levels. A list of product attributes used in the survey is presented in Table 1. There are six brands comprising the set of inside goods and a total of six different attributes. The first column of attribute levels is listed parenthetically to indicate the reference levels for each attribute. Part-worths for the remaining attribute levels represent the marginal utility of the indicated level relative to the reference level for each attribute.

The conjoint data were collected using a dual-response format with six alternatives, followed by a buy/no-buy task to indicate whether the respondent would purchase the preferred television. In each choice task, respondents were presented with six different

Table 1 Television study product attributes and levels

Attribute	Levels					
	1	2	3	4	5	6
Brand	Sony	LG	Visio	Hisense	Samsung	TCL
Screen size	(55")	65"	75"	—	—	—
Resolution	(4K)	8K	—	—	—	—
Refresh	(60Hz)	120Hz	—	—	—	—
Display type	(LED)	QLED	OLED	—	—	—
Premium processing & sound	(No)	Yes	—	—	—	—
Price	\$599	\$999	\$1,399	\$1,799	\$2,199	—

Note: The attribute levels in the parentheses are used as the null level of each corresponding attribute

TV options that varied across key attributes including brand, screen size, resolution, refresh rate, display type, premium processing and sound, and price. Respondents were first asked to select the TV they preferred the most from the six options. Then, in a follow-up question, they were asked whether they would actually purchase the TV they had selected. Each respondent completed 15 choice tasks in the survey, and a total of 318 qualified respondents participated, providing information about their preferences. Eight respondents who chose not to purchase in all 15 choice tasks were removed because they do not provide reliable information about part-worths of product features nor their sensitivity to prices, resulting in a final sample of 310 respondents for analysis. Retaining these individuals leads to inflated estimates of equilibrium prices reported below.

To analyze the data using the Standard Model, preference information for the inside goods was excluded if the no-choice option was selected (i.e., if the respondent answered “No” to “Would you purchase the preferred TV selected above?”). This data set is referred to as the single-response data in the following section.

4 Results

We compare the performance of our proposed dual-response Unified Model to the Standard Model with and without budget constraints. The specification of the Standard Model is provided in Eq. 1, the Unified Model without a budget constraint is specified by Eq. 4, and the Unified Model with a budget constraint on the second response task is specified by Eq. 5. Price is scaled in terms of \$100’s of dollars, and the estimated part-worth utilities are measured relative to the reference levels. All models are estimated as hierarchical Bayesian models that allow for respondent heterogeneity, utilizing diffuse prior settings. For all models, we ran 400,000 iterations with a burn-in of the first 200,000 draws and thinning every tenth draw.

4.1 Model fit

We measure the fit of the single and dual-response models using the log marginal density (LMD) statistic, employing the Newton-Raftery (Newton and Raftery, 1994) approximation. Table 2 reports the LMD for single-response data, and Table 3 reports the LMD for dual-response data. The results show that the budget constraint improves the fit of both models and that the budget constraint on the second response task only fits best for dual-response data. Note that the LMD for the models using the single-response data and the models using the dual-response data are not directly comparable

Table 2 Model fit using the single-response data

Model	LMD
Standard model without budget constraint	−4524
Standard model with budget constraint	−4418

Table 3 Model fit using the dual-response data

Model	LMD
Unified model without budget constraint	−5961
Unified model with budget constraint on both	−5939
Unified model with budget constraint on second	−5735

because they are based on different data sets, and we will compare these models in further detail in Sect. 5.1.

4.2 Parameter estimates

Table 4 reports the mean of the random effects for the attribute part-worths, price coefficient, and budgetary allotment ($\bar{\phi}$). We find general agreement in the part-worth estimates for the Standard (single-response) and Unified (dual-response) models without budget constraints. The parameter estimates reported are within two posterior standard deviations of each other, indicating that both single-response and dual-response data lead to consistent estimates of the true data-generating process. For example, the part-worth estimate for 8K resolution is estimated to be about 0.50 in all models. Furthermore, estimated market shares for the Standard and Unified models without budget constraints result in nearly identical choice shares.

When budget constraints are included in the Standard Model, we find that the price coefficient decreases. This is because the Standard Model without the budget

Table 4 Posterior mean (standard deviation) of random effect ($\bar{\phi}$)

Budget constraint	Standard Model (on Single-Response Data)		Unified Model (on Dual-Response Data)		
	×	✓	×	✓ (on both)	✓ (on second)
Sony	1.43 (0.29)	0.63 (0.27)	1.00 (0.29)	1.04 (0.28)	1.39 (0.28)
LG	1.42 (0.29)	0.60 (0.27)	0.91 (0.28)	0.94 (0.28)	1.31 (0.28)
Visio	0.60 (0.31)	−0.20 (0.29)	0.33 (0.29)	0.37 (0.29)	0.72 (0.29)
Hisense	−0.02 (0.32)	−0.83 (0.31)	−0.30 (0.29)	−0.27 (0.30)	0.10 (0.29)
Samsung	1.89 (0.29)	1.19 (0.26)	1.37 (0.28)	1.42 (0.28)	1.76 (0.28)
TCL	0.28 (0.32)	−0.50 (0.29)	0.02 (0.30)	0.06 (0.29)	0.41 (0.29)
65inch	0.53 (0.10)	0.60 (0.10)	0.51 (0.08)	0.51 (0.08)	0.50 (0.08)
75inch	0.86 (0.12)	0.94 (0.13)	0.78 (0.11)	0.79 (0.11)	0.78 (0.10)
8K	0.58 (0.09)	0.57 (0.09)	0.45 (0.08)	0.45 (0.08)	0.43 (0.07)
120Hz	0.60 (0.08)	0.63 (0.09)	0.57 (0.07)	0.58 (0.07)	0.56 (0.07)
QLED	0.42 (0.10)	0.37 (0.10)	0.30 (0.09)	0.30 (0.09)	0.32 (0.08)
OLED	0.70 (0.11)	0.71 (0.10)	0.58 (0.09)	0.58 (0.09)	0.59 (0.09)
Sound-Yes	0.78 (0.10)	0.79 (0.10)	0.71 (0.09)	0.71 (0.09)	0.70 (0.09)
Price (β_p^*)	−1.91 (0.11)	−2.22 (0.12)	−2.17 (0.10)	−2.16 (0.10)	−2.29 (0.11)
Budget		24.06 (1.10)		26.77 (1.30)	23.92 (0.76)

constraint needs to have a larger marginal price coefficient to model the fact that the respondents are not choosing the higher-priced offerings. The price coefficient does not change when the budget constraint is applied to the second response task in the Unified Model because the information about price sensitivity comes primarily from the first response and not the second in the dual-response data.

5 Discussion

The advantage of using a dual-response format in conjoint analysis is that it increases the amount of information in the data by capturing an additional response and isolating the effect of the budget constraint in the second no-choice decision. In this section, we examine the information gain when using the Unified Model with the dual-response format compared to the Standard Model with the single-response format. In addition, we further investigate the effect of the budget constraint on optimal pricing.

5.1 Information gain

The Fisher measure of information is widely used to determine the optimal design of conjoint choice experiments (Huber and Zwerina, 1996; Kanninen, 2002; Sándor and Wedel, 2001, 2005; Kessels et al., 2006, 2009). We use this measure to compare the gain in efficiency when employing the Unified Model for dual-response formatted data relative to the Standard Model for single-response data. Note that we cannot directly compare the LMDs of the Standard Model in Table 2 with those of the Unified Model in Table 3 due to the different data formats. However, the Fisher information represents the precision of the model estimates, which allows for a meaningful comparison between the Standard Model and the Unified Model, as they share the same model parameters.

We begin with the design matrix for the response task t , given by $X_t = [\tilde{x}_{t1}, \dots, \tilde{x}_{tJ}]'$, where $\tilde{x}_{tj} = (x_{tj}, p_{tj})'$ is the vector of attribute levels and the price of the alternative j in the response task t . Let $V_{tj} = \beta'x_{tj} + \beta_p p_{tj}$ the deterministic part of utility from selecting alternative j in response task t and

$$\begin{aligned} P_t &= \left[\frac{\exp(V_{t1})}{1 + \sum_{k=1}^J \exp(V_{tk})}, \dots, \frac{\exp(V_{tJ})}{1 + \sum_{k=1}^J \exp(V_{tk})} \right]', \\ P_t^* &= \left[\frac{\exp(V_{t1})}{\sum_{k=1}^J \exp(V_{tk})}, \dots, \frac{\exp(V_{tJ})}{\sum_{k=1}^J \exp(V_{tk})} \right]', \\ P_{t,nc} &= \frac{1}{1 + \sum_{k=1}^J \exp(V_{tk})}. \end{aligned}$$

The Fisher information matrix, which is the inverse of the asymptotic variance-covariance matrix of the model parameter estimates (McFadden, 1974), is obtained as the variance of the first-order derivatives of the log-likelihood function with respect to the model parameters. With a detailed derivation provided in the Appendix, the

Fisher information matrix $I(X, \beta, \beta_p)$ for the Standard Model is given by Huber and Zwerina (1996):

$$I(X, \beta, \beta_p) = \sum_{t=1}^T [X_t' \text{diag}(P_t) X_t - X_t' P_t P_t' X_t].$$

and the Fisher information matrix $\tilde{I}(X, \beta, \beta_p)$ for the Unified Model extends the Fisher information matrix for the Standard Model by incorporating an additional term:

$$\tilde{I}(X, \beta, \beta_p) = I(X, \beta, \beta_p) + \sum_{t=1}^T P_{t,nc} \left(X_t' \text{diag}(P_t^*) X_t - X_t' P_t^* P_t^{*'} X_t \right).$$

A widely accepted one-dimensional measure of efficiency in conjoint choice experiment designs, the $\mathcal{D}$ -criterion value, is used here to compare the Fisher information matrices of the Standard Model and the Unified Model (Huber and Zwerina, 1996; Sándor and Wedel, 2001; Kessels et al., 2006). The $\mathcal{D}$ -criterion value is defined as the determinant of the inverse of the information matrix:

$$\mathcal{D} = \{\det I(X, \beta, \beta_p)^{-1}\}^{1/k}$$

where $I(X, \beta, \beta_p)$ is the Fisher information matrix with respect to the design matrix X and a certain k -dimensional model parameter $(\beta, \beta_p)'$. The exponent $1/k$ normalizes the determinant of the information matrix, ensuring it remains independent of the dimensionality k of the model parameter. A lower $\mathcal{D}$ -criterion value is preferred, as it corresponds to a smaller determinant of the variance-covariance matrix of the parameter estimators, $\{\det I(X, \beta, \beta_p)^{-1}\}$.

For a Bayesian approach, the $\mathcal{D}_{\mathcal{B}}$ -criterion value is obtained by integrating $\mathcal{D}$ -criterion value over the posterior distribution of model parameters (Sándor and Wedel, 2001, 2005; Kessels et al., 2006, 2009):

$$\mathcal{D}_{\mathcal{B}} = \mathbb{E}_{(\beta, \beta_p)'} [\{\det I(X, \beta, \beta_p)^{-1}\}^{1/k}] = \int \{\det I(X, \beta, \beta_p)^{-1}\}^{1/k} \pi(\beta, \beta_p) d\beta d\beta_p.$$

We use the last 100 posterior draws from the posterior distribution of each respondent's part-worths to investigate the information gain when using the Unified Model for dual-response data.

For each fixed number of response tasks m , we compute the smallest $\mathcal{D}_{\mathcal{B}}$ -criterion value among all possible combinations of m response tasks selected from the 15 tasks for the Standard Model and the Unified Model, respectively. This allows us to identify the most efficient subset of response tasks, as measured by the $\mathcal{D}_{\mathcal{B}}$ -criterion value for both models, and to compare their efficiency based on these optimal subsets.

We calculate the best $\mathcal{D}_{\mathcal{B}}$ -criterion values for a number of tasks ranging from four to 15. Values for fewer than three tasks are not calculated because some combinations result in a singular information matrix, making it impossible to compute its inverse.

For each number of tasks, the Unified Model consistently achieves lower $\mathcal{D}_{\mathcal{B}}$ -criterion values compared to the Standard Model, with an average information gain of approximately 30%. Notably, the Unified Model with only 10 tasks already achieves a better $\mathcal{D}_{\mathcal{B}}$ -criterion value than the Standard Model does with 15 tasks.

5.2 Optimal pricing

The presence of the budget constraint has implications for optimal pricing. We investigate optimal pricing by comparing Nash equilibrium prices with and without the budget constraint. The standard static Nash equilibrium is a set of prices that simultaneously satisfy the profit-maximization conditions of all offerings, which is used as a counterfactual outcome in conjoint experiments (Choi and DeSarbo, 1994; Srinivasan et al., 1997; Allenby et al., 2014; Pachali et al., 2023).

To reach a Nash equilibrium, each manufacturer, one at a time, determines the optimal prices of their offerings given the prices of other offerings and their costs. This optimization is repeated for each manufacturer until an equilibrium is reached where no brand has an incentive to change prices further. The equilibrium is referred to as a Nash equilibrium because none of the brands has an incentive to deviate from their profit-maximizing price, given the prices of the other brands.

In a choice setting, the profit function for brand j is given by

$$\pi(p_j \mid p_{-j}) = MS \times \mathbb{E}[\Pr(j \mid p, A)](p_j - c_j), \quad (8)$$

where MS is the market size, $\Pr(j \mid p, A)$ is the probability of choosing brand j given the price vector of all J brands in the market, $p = (p_1, \dots, p_J)$, A is the current choice set of competitive offerings in the market, and c_j is the marginal cost of producing brand j 's product. We approximate the market shares as the expected choice probability computed via Monte Carlo draws from the posterior distribution, where the expectation is taken with respect to the posterior distribution of the choice model parameters. Details are provided in the Appendix.

$$\mathbb{E}[\Pr(j \mid p, A)] = \int \Pr(j \mid p, A) \pi(\beta, \beta_p^*) d\beta d\beta_p^*,$$

for the model without the budget constraint, and

$$\mathbb{E}[\Pr(j \mid p, A)] = \int \Pr(j \mid p, A) \pi(\beta, \beta_p^*, M) d\beta d\beta_p^* dM,$$

for the model with the budget constraint.

The market scenario we assume includes six TV options that are identical in all product attributes except brand, with the inclusion of an outside-good option. Each TV features a 75" screen size, 4K resolution, 120Hz refresh rate, OLED display technology, and premium processing and sound. The only difference between the options is the brand, which includes LG, Visio, Samsung, Hisense, TCL, and Sony.

We assume that the marginal cost of producing each TV is \$1,000 for all brands and use the Standard Model and the Unified Model with a budget constraint on the second response task to illustrate the influence of the budget constraint.

The Nash equilibrium prices differ significantly between the Standard Model and the Unified Model. In the Standard Model, the equilibrium prices are \$3,461 for LG, \$2,264 for Visio, \$4,065 for Samsung, \$2,932 for Hisense, \$2,035 for TCL, and \$4,271 for Sony. In the Unified Model, which incorporates a budget constraint on the second response task, the corresponding prices drop to \$1,953 for LG, \$1,688 for Visio, \$2,197 for Samsung, \$2,199 for Hisense, \$1,597 for TCL, and \$2,151 for Sony.

The equilibrium prices in the Standard Model appear to be misleading, as the optimal prices for LG, Samsung, Hisense, and Sony are extremely high. This occurs because the Standard Model does not restrict the choice probability when the price of an alternative exceeds the budget constraint. Although the probability of choosing a product decreases as its price increases, the change in market share is small enough that a price increase still results in higher profit. In contrast, our Unified Model, which incorporates a budget constraint on the second response task, ensures that respondents do not consider products if the price exceeds their budget, effectively removing them from the market.

6 Conclusion

The results of our analysis demonstrate the advantage of using a dual-response format in conjoint analysis in providing more efficient parameter estimates and disentangling the budget constraint from the first choice task. We find that respondents do not consider their budget when responding to the first choice task, allowing them to indicate their preference for all choice alternatives. The effect of a respondent's budget is revealed in their response to the second response task. Even when all prices in the choice set fall within respondents' budgets, the determinant of the Fisher information matrix from the Unified Model for dual responses is larger than that of the Standard Model for single responses, indicating that using the Unified Model is more informative.

We find that part-worth estimates for dual and single responses are approximately the same but diverge when budget constraints are added to the model. Models that do not account for the budget tended to overstate price sensitivity in the Standard Model. By incorporating the dual-response format and budget constraints, the proposed model provides more precise estimates of preferences and more realistic pricing implications.

We demonstrate the influence of budget constraints by investigating the optimal prices in a TV market using a Nash price equilibrium analysis. The Standard Model overstates the equilibrium price of TV for Samsung by 85.0%, and the equilibrium price of TV for LG by 77.2%, respectively. These findings have significant implications for determining optimal prices from a conjoint study. Overall, our results support the use of a dual-response format in conjoint analysis with a budget constraint added to the second response.

Appendix A: Fisher information matrix for the standard model and the unified model

In this appendix, we present the information matrix for the Standard Model and the Unified Model.

the standard model

Let $\tilde{x}_{tj} = (x'_{tj}, p_{tj})'$, where x_{tj} is the vector containing the attribute levels and p_{tj} is the price for alternative j in response task t . Similarly, let $\tilde{\beta} = (\beta, \beta_p)'$, where β represents the part-worths for the corresponding attribute levels, and β_p denotes the marginal utility of money.

Then, with a total of J inside-good options in task t , the probability of choosing alternative j is given by

$$P_{tj} = \frac{\exp(\tilde{\beta}'\tilde{x}_{tj})}{\sum_{k=0}^J \exp(\tilde{\beta}'\tilde{x}_{tk})}, \quad j = 0, \dots, J$$

where we denote $x_{t0} = 0$ for the outside-good option. For simplicity, we remove the tilde symbol from β and x_{tj} in the following sections.

The log-likelihood function for the response task t is given by

$$\log(L_t) = \sum_{j=0}^J y_{tj} \log(P_{tj}),$$

where $y_{tj} = 1$ if alternative j is chosen in task t , and $y_{tj} = 0$ otherwise.

Differentiating the log-likelihood function with respect to β and further differentiating the derivative, we obtain

$$\begin{aligned} \frac{\partial \log(L_t)}{\partial \beta} &= \sum_{j=0}^J \frac{y_{tj}}{P_{tj}} \frac{\partial P_{tj}}{\partial \beta}, \\ \frac{\partial^2 \log(L_t)}{\partial \beta \partial \beta'} &= \sum_{j=0}^J \frac{y_{tj}}{P_{tj}} \left[\frac{\partial^2 P_{tj}}{\partial \beta \partial \beta'} - \frac{1}{P_{tj}} \frac{\partial P_{tj}}{\partial \beta} \frac{\partial P_{tj}}{\partial \beta'} \right]. \end{aligned}$$

Next, we have

$$\frac{\partial P_{tj}}{\partial \beta} = P_{tj} (x_{tj} - \sum_{k=0}^J x_{tk} P_{tk}) = P_{tj} (x_{tj} - \bar{x}_t),$$

where $\bar{x}_t = \sum_{k=0}^J x_{tk} P_{tk}$.

Also, we have

$$\frac{\partial^2 P_{tj}}{\partial \beta \partial \beta'} = P_{tj} (x_{tj} - \bar{x}_t)(x_{tj} - \bar{x}_t)' - P_{tj} \sum_{k=0}^J P_{tk} x_{tk} x'_{tk} + P_{tj} \bar{x}_t \bar{x}_t'$$

By inserting the equations above, we obtain

$$\begin{aligned}
 \frac{\partial^2 \log(L_t)}{\partial \beta \partial \beta'} &= \sum_{j=0}^J \frac{y_{tj}}{P_{tj}} \left[P_{tj} (x_{tj} - \bar{x}_t) (x_{tj} - \bar{x}_t)' - P_{tj} \sum_{k=0}^J P_{tk} x_{tk} x_{tk}' + P_{tj} \bar{x}_t \bar{x}_t' \right. \\
 &\quad \left. - P_{tj} (x_{tj} - \bar{x}_t) (x_{tj} - \bar{x}_t)' \right] \\
 &= \left[\bar{x}_t \bar{x}_t' - \sum_{k=0}^J P_{tk} x_{tk} x_{tk}' \right] \left(\sum_{j=0}^J y_{tj} \right) \\
 &= \bar{x}_t \bar{x}_t' - \sum_{k=0}^J P_{tk} x_{tk} x_{tk}'
 \end{aligned}$$

Let

$$\begin{aligned}
 X_t &= [x_{t1}, \dots, x_{tJ}]', \\
 P_t &= (P_{t1}, \dots, P_{tJ})',
 \end{aligned}$$

and by the fact that

$$\bar{x}_t = \sum_{k=0}^J x_{tk} P_{tk} = \sum_{k=1}^J x_{tk} P_{tk} = X_t' P_t,$$

we obtain the information matrix I for the Standard Model:

$$\begin{aligned}
 I(X, \beta) &= \sum_{t=1}^T \mathbb{E} \left[- \frac{\partial^2 \log(L_t)}{\partial \beta \partial \beta'} \right] \\
 &= \sum_{t=1}^T [X_t' \text{diag}(P_t) X_t - X_t' P_t P_t' X_t]
 \end{aligned}$$

the Unified Model

Let x_{tj} and β defined the same as in the last section. Then, with a total of J inside-good options in response task t , let

$$\begin{aligned}
 y_{tj} &= \begin{cases} 1, & \text{if alternative } j \text{ is chosen in the first task,} \\ 0, & \text{otherwise.} \end{cases} \\
 y_{t,nc} &= \begin{cases} 1, & \text{if "not to purchase" in the second task,} \\ 0, & \text{otherwise.} \end{cases} \\
 W_t &= \sum_{k=1}^J \exp(\beta' x_{tk})
 \end{aligned}$$

$$P_{tj}^* = \frac{\exp(\beta' x_{tj})}{\sum_{k=1}^J \exp(\beta' x_{tk})}, \quad j = 1, \dots, J$$

$$P_t^* = (P_{t1}^*, \dots, P_{tJ}^*)'$$

we have the log-likelihood function for response task t ,

$$\log(\tilde{L}_t) = \sum_{j=1}^J y_{tj} \log(P_{tj}) - y_{t,nc} \log(W_t)$$

By differentiating the log-likelihood function with respect to β and further differentiating the derivative, we obtain

$$\frac{\partial \log(\tilde{L}_t)}{\partial \beta} = \sum_{j=1}^J \frac{y_{tj}}{P_{tj}} \frac{\partial P_{tj}}{\partial \beta} - \frac{y_{t,nc}}{W_t} \frac{\partial W_t}{\partial \beta},$$

$$\frac{\partial^2 \log(\tilde{L}_t)}{\partial \beta \partial \beta'} = \sum_{j=1}^J \frac{y_{tj}}{P_{tj}} \left[\frac{\partial^2 P_{tj}}{\partial \beta \partial \beta'} - \frac{1}{P_{tj}} \frac{\partial P_{tj}}{\partial \beta} \frac{\partial P_{tj}}{\partial \beta'} \right] - \frac{y_{t,nc}}{W_t} \left[\frac{\partial^2 W}{\partial \beta \partial \beta'} - \frac{1}{W_t} \frac{\partial W_t}{\partial \beta} \frac{\partial W_t}{\partial \beta'} \right].$$

We have

$$\frac{\partial W_t}{\partial \beta} = \sum_{k=1}^J x_{tk} \exp(\beta' x_{tk}) = W_t \sum_{k=1}^J P_{tj}^* x_{tk}$$

$$\frac{\partial^2 W_t}{\partial \beta \partial \beta'} = \sum_{k=1}^J x_{tk} x_{tk}' \exp(\beta' x_{tk}) = W_t \sum_{k=1}^J P_{tj}^* x_{tk} x_{tk}'$$

By inserting the equations above, we have

$$\frac{\partial^2 \log(\tilde{L}_t)}{\partial \beta \partial \beta'} = \bar{x}_t \bar{x}_t' - \sum_{k=0}^J P_{tk} x_{tk} x_{tk}' - y_{t,nc} \left[\sum_{k=1}^J P_{tj}^* x_{tk} x_{tk}' - \left(\sum_{k=1}^J P_{tj}^* x_{tk} \right) \left(\sum_{k=1}^J P_{tj}^* x_{tk} \right)' \right]$$

and the information matrix $\tilde{I}$ for the Unified Model:

$$\tilde{I}(X, \beta) = \sum_{t=1}^T \mathbb{E} \left[- \frac{\partial^2 \log(\tilde{L}_t)}{\partial \beta \partial \beta'} \right]$$

$$= I(X, \beta) + \sum_{t=1}^T \{ p_{t,nc} (X_t' \text{diag}(P_t^*) X_t - X_t' P_t^* P_t^{*'} X_t) \},$$

where

$$p_{t,nc} = \frac{1}{1 + \sum_{k=1}^J \exp(\beta x_{tk})}.$$

Appendix B: Market shares approximation in the optimal pricing

We approximate the expected market share using the posterior draws of each individual's part-worths. Let H be the total number of respondents and Itr the total number of posterior draws. Define x_j and p_j as the attribute levels and price for alternative j in the current choice set of competitive offerings in the market, where $j = 1, \dots, J$, with a total of J alternatives. Let $(\beta_h^{(d)}, \beta_{ph}^{*(d)})$ denote the d -th draw of the individual posterior part-worths, and $M_h^{(d)}$ the d -th draw of the budget for respondent h . For the model without the budget constraint, the expected market share is given by:

$$\mathbb{E}[\Pr(j \mid p, A)] \approx \frac{1}{H \times Itr} \sum_{d=1}^{Itr} \sum_{h=1}^H \frac{\exp(\beta_h^{(d)'} x_j - \beta_{ph}^{(d)} p_j)}{\sum_{k=1}^J \exp(\beta_h^{(d)'} x_k - \beta_{ph}^{(d)} p_k)}.$$

And for the model with the budget constraint, the expected market share is:

$$\mathbb{E}[\Pr(j \mid p, A)] \approx \frac{1}{H \times Itr} \sum_{d=1}^{Itr} \sum_{h=1}^H \frac{\exp(\beta_h^{(d)'} x_j - \beta_{ph}^{(d)} p_j) \mathbb{1}(p_j \leq M_h^{(d)})}{\sum_{k=1}^J \exp(\beta_h^{(d)'} x_k - \beta_{ph}^{(d)} p_k)}.$$

where $\beta_{ph}^{(d)} = -\exp[\beta_{ph}^{*(d)}]$.

Data Availability The data can be provided upon request.

Declarations

Ethical approval The data collection for this study has been approved by the Institutional Review Board (IRB) of the institution (Study Number: 2023B0019), along with a waiver of consent documentation.

Conflict of interest The authors declare no competing interests.

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